

Technische Hochschule Ingolstadt

Fakultät Informatik

Augmented and Virtual Reality Applications Practical Exercises

Exercise 03: Marker-based Tracking in AR

Introduction to AR

Overview

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- Task 2: Create new Unity project
- Task 3: Import Vuforia tracking-target image and 3D model
- Task 4: Photometric registration
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Exercise 03: Marker-based Tracking in AR (Introduction to Vuforia in Unity)

In this exercise, we'll create a handheld AR Application with "Vuforia" and Unity.

"Vuforia" is a AR software platform widely used for industrial applications like manufacturing and maintenance, as well as in marketing, training, and education. It recognizes predefined tracking markers and allows geometric registration based on them. We want to augment reality - captured by a webcam/smartphone/tablet camera - with a virtual object (a 3D-model of a motor engine block).

This is the first part (of two) using "Vuforia" in Unity. Part two will be the next mandatory assignment.

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Task 1: Set up a new Vuforia account

a) To use the Vuforia framework in Unity, we first need a Vuforia account (free Developer Account).
Go to the Vuforia developer *website* and create a new account (upper right → Register).

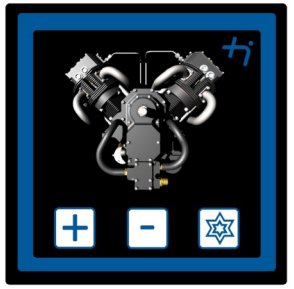
b) After logging in, we need to create a new project with a valid license key:
“My Account” (upper right) → “Plan & Licenses” tab, “Generate Basic License”.

c) Give your license any name and confirm/agree/check terms and conditions of the “Vuforia Developer Agreement”.

d) Back in “Plan & Licenses”, click on the newly created license in the list to view the “License Key”.
We will need this later when using Vuforia in Unity.

e) To use the tracking-marker target images in our project, we first need to create a new target database containing our tracking-marker images that we’ll use later:
“Target Manager” → “Generate Database”. Rename it (e.g., "THI"), type “Device”.

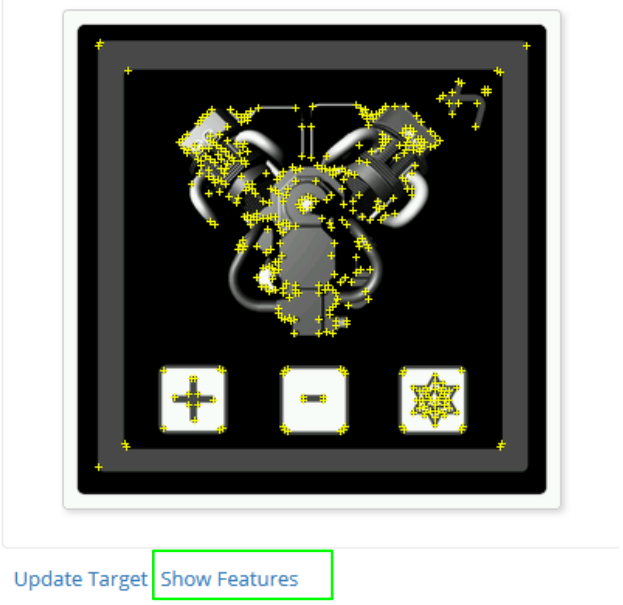
f) Download **EngineMarker.png** from *Moodle*. This will be our marker where we’ll place a virtual object onto:



Task 1: Set up a new Vuforia account

g) On the Vuforia website, click on the newly created database: “Add Target” → Type: “Image”. Upload **EngineMarker.png** and specify the width as 0.2 (this corresponds to 20 cm in Unity’s real-world units). Click “Add” to add the new tracking target to the database.

The rating-stars tell us how augmentable it is, i.e. how well an image is suited as a tracking marker. The engine marker is a very good marker: click on “enginemarker”, and then “Show Features”. This shows us the detected features (distinct points recognized in the image). Try to understand by which characteristics in the image these tracking-features are extracted and defined.



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Task 1: Set up a new Vuforia account

h) In the Target Manager Database, click “Download Database (All)”. Development Platform: “Unity Editor”. Download.
This will create a .unitypackage file.
We’ll need to import this Unity package later to use this newly created tracking target.



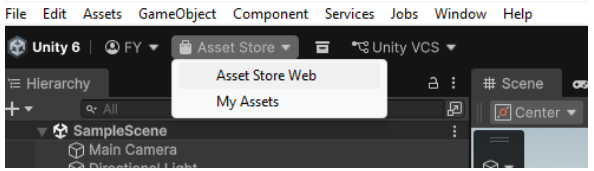
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Task 2: Set up a new Unity project

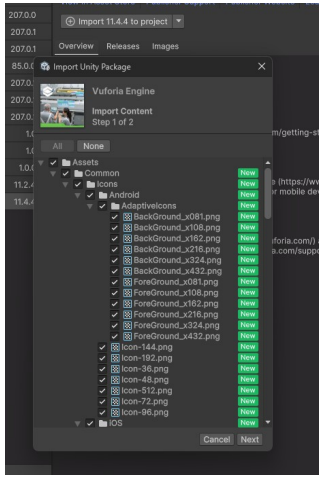
a) Launch Unity Hub. Create a New Project with Version 6.000.3..11f1. Type: Universal 3D **SRP** (as of Oct. 2026, only **S**tandard **R**ender **P**ipeline is supported, URP (Universal Render Pipeline) is not).

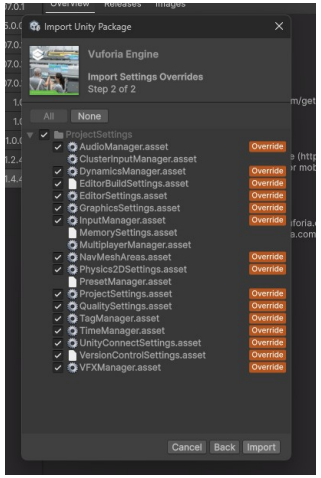
b) In Unity: Menu → Window → Package Management → Asset Store. Or, easier:



Both ways open https://assetstore.unity.com.
Search for “Vuforia Engine SDK”. Select it, “Add to my Assets” (no confirmation).
This step doesn’t yet bring the Engine into our project.
It only “adds/transfers/links” it to our Unity Account. We need to import it into our project:

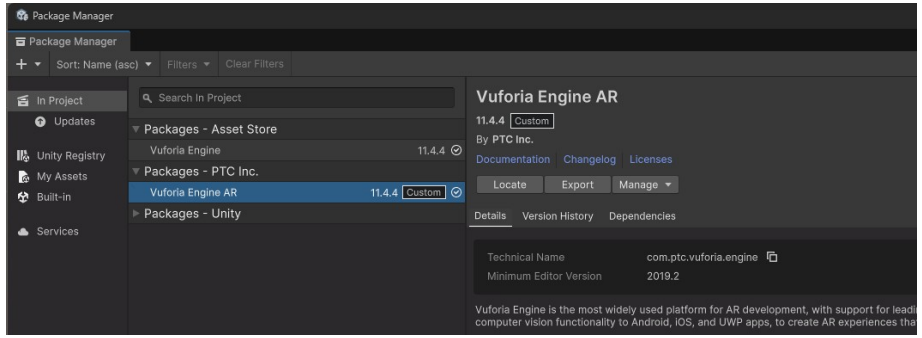
c) Menu → Window → Package Management →Package Manager.
Under “My Assets”, you should see “Vuforia Engine” in the list: → “Import 11.4.4. to Project”.
Warning! → “Import”. Warning. → “Install/Upgrade”. Wait.
PopUp (“Import Unity Package”):→ “Next”. Overrides: → “Import”. Wait.
Warning. → “Update”. Wait. Missing Signature! → “Open Package Manager”. Warning.
Dropdown: “Manage” →”Customize”. Wait. Done.
Easy.





Task 2: Set up a new Unity project

Make sure it's imported correctly:
Still in the Package Manager, under "My Project": Version 11.4.4. (September 09 2025):



d) Close the Package Manager. In Menu → GameObject, there's now a new sub-menu called **Vuforia Engine**.
Create an **AR Camera**. We now don't need the **Main Camera** any more. Delete.

e) We'll now import the **Vuforia Database Unity Package** for the tracking image target we had downloaded from Vuforia in task 1h) :
Menu → Assets → Import Package → Custom Package. Locate **[yourVuforiaDatabaseName].unitypackage** and import.

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Task 2: Set up a new Unity project

e) Select the **ARCamera** object in the Hierarchy. Inspector → Vuforia Behaviour Script → click “Open Vuforia Engine Configuration”.

Copy paste your Vuforia license key (long cryptic string) from the Vuforia database into the “App License Key” textbox.
Next, to activate the webcam (either your built-in, or the webcam connected to the workstation), scroll down to “Play Mode”. Set “PlayMode Type” to “WEBCAM” and select your “Camera Device” model:



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Task 3: Add Image Target and 3D Model in Unity

a) In order to use our tracking marker in Unity, we fneed to add a **Vuforia ImageTarget** to the scene:
Add a new GameObject → Vuforia Engine → ImageTarget.

b) Inspector → Image Target Behaviour → Type: “From Database”.
Select your database from the drop-down menu. “EngineMarker.png” image should now appear in “ImageTarget”.
Quick check:
Select the “Image Target” Game Object in the Hierarchy Pane, and hit F (the Viewport zooms onto the EngineMarker Plane).

c) Let’s import our virtual AR object. We could use any 3D-Model - here`s one already prepared for this exercise:
download **V-Engine.fbx** from *Moodle*. Drag and drop the file into the “Assets” folder Unity’s Project pane.

d) Attach and position the Model: Drag the V-Engine prefab into the Hierarchy as a child of **ImageTarget.** Scale it down to 0.05 in the Inspector. Move the model slightly upwards along the Y-axis so it floats above the marker. Think about the Y-Position in real-world-units.

e) Test the application: First, transfer the tracking-marker image to your smartphone (or print it out on paper).
Save your scene. Enter Play-Mode in Unity and hold the image in front of your webcam/laptop cam: the 3D engine model should appear floating above the marker.
Discussion: Why would printing the tracking marker on paper provide a more reliable tracking experience than displaying it on a smartphone screen?

Task 4: Photometric Registration

a) Once geometric registration works, let`s do photometric registration as well (remember the lecture).
The motor block doesn’t look very realistic, neither does the lighting. In order to make it integrate better into the real Lab-Environment: try to mimic the real light sources (position, color, intensity). Do not change the materials of the 3D-model itself.
Adjust the lights one by one: trial-and-error. Manual work….

That’s it with this exercise. If you still have some time and/or motivation: next slide (optional).

Task 5 - optional: Build app for Android tablet/smartphone
If you want, you can build the app for an Android tablet or smartphone.
Disclaimer: I don't own any Android device, so I couldn't test this part of the exercise myself. No warranties. This is how it *should work in theory.*

a) Menu → File → Build Profiles (or Ctrl Shift B).
Select the Platform (Android), press "Open Scene List", "Add Open Scenes", and select your scene in the list (it should have a "1" after). If you only have one scene (Project Pane -> Assets → Scenes), it's selected by default.

b) Select "Android" as the platform and press "Switch Platform". Takes some while. Press "Build". Choose a location & filename for the .apk installation file. Wait.

c) Transfer the .apk to your Android smartphone (or tablet).

d) Your smartphone must be in developer mode to install the .apk file. When installing, you will see a warning and must confirm that you trust the software.

e) Print out the image marker, put it on the table and film it with your Android device. The engine model appears.

End of exercise 03.